

Poverty and Inequality Research Lab
Research Assistant - Predoctoral Fellow Program Data Exercise
2026 Application Cycle

Your performance on this exercise will be used to evaluate your ability to leverage data to answer research questions. The data work itself is not meant to be terribly difficult or advanced but does presuppose some knowledge of basic multi-variate regression analysis. We are most interested in the degree to which you understand estimated coefficients, including questions of correlation versus causality and, more generally, how you relate econometric or other types of data analysis to real-world, policy-relevant issues. We will also be looking for careful thoughtful work, cogent writing, cleanly labeled and formatted tables, and clear explanations.

To complete the exercise, you should do the following:

1. Upload the data in this folder to a statistical analysis program, such as STATA, that can be used to run basic regressions. There are four variables: country, income, changed_behaviors, and outside_space. The latter two are binary variables, the first indicating if the individual changed their behavior in an effort to avoid infection from Covid-19 and the second indicating if the person has access to outside space in their home. The income variable is categorical rather than continuous. Please note that the data are in stata “.dta” and also “.csv” formats.
2. Once you’ve uploaded the data, please choose one country to study and drop observations from other countries. Next, please make a table (call it Table 1) that describes the data, i.e., presents summary statistics. If you are not sure how to do this, please go to Prof. Papageorge’s website, find a published research paper and see how he presents basic details of data sets. You should describe the data set in general and the summary statistics. Most of the information about the full data set is found in a paper on Papageorge’s website.¹
3. Next, relate behavior change to having access to outside space at home. There are different ways this regression can be run. Please be very clear about which specific regression you’ve run and discuss alternative specifications and ways of using the variables, given how they were created. Please make a well-formatted table that presents your results (call it Table 2).
4. Discuss your results from Step 3. In particular, focus on how to interpret the estimated relationship. You should probably discuss correlation versus causality, omitted variables (and omitted variables bias), endogeneity and related concepts. What would be an ideal experiment that would allow you to make causal claims? How does your analysis depart from that ideal experiment? What are some key threats to identification? How might one get around them? What is the direction of bias? Why? Are there credible IV’s you might search for?
5. Next, incorporate income into your analysis. What do you find? What is your interpretation? How does this affect the coefficient estimates from Step 3? How does the addition of

¹ A link to the document is here: <https://docs.iza.org/dp13230.pdf>

income relate to your discussion of endogeneity and omitted variables bias? Please also add the income regression results to Table 2.

6. Suppose the “outside space at home” results could be assigned a causal interpretation. Please discuss policy implications and future work. In particular:
 - a. What sorts of short-run or long-run policies or programs should we consider? Why might they be effective in light of your results?
 - b. If you could collect additional quantitative data to get a better sense of the empirical results, what data would you try to collect? What additional variables you would like to collect? Explain why you would want them and how would you use them to augment your analysis.
7. Suppose you could collect qualitative data. How would you do that? Consider the possibility of open-ended interviews. What questions would you ask and why? How would you sample households or individuals? How could answers to those questions augment your analysis? What might you gain from the qualitative analysis that you might be unable to learn from additional quantitative data collection? Would you also include ethnographic observations? What value might those observations have beyond what you could learn from interviews? How might your answer to this question affect your answer to Step 6? If you are not sure how to approach this, please go to the PIRL website and look at one of Prof. DeLuca’s published research papers and see how she describes qualitative data design and analysis.
8. Once you are finished, please submit your work as a cleanly formatted document that answers questions and presents results. In total, you should not have more than about 2-3 pages of writing and an additional page for tables. Please also include “do-files” or other documentation of the analyses you’ve run.